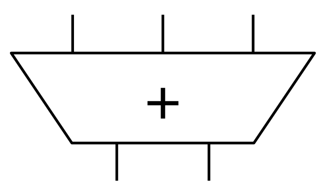
Cycle 0

Cycle

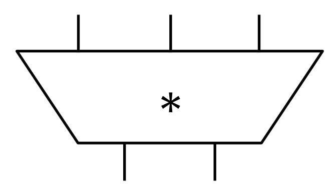
```
MUL  R1, R2  → R3
ADD  R3, R4  → R5
ADD  R2, R6  → R7
ADD  R8, R9  → R10
MUL  R7, R10 → R11
ADD  R5, R11 → R5
```

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		3
R4	1		4
R5	1		5
R6	1		6
R7	1		7
R8	1		8
R9	1		9
R10	1		10
R11	1		11

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a						
b						
c						
d						



	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x						
y						
z						
t						

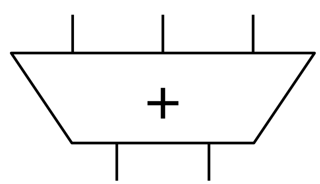


Cycle 1

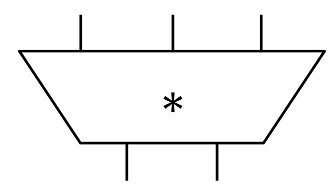
	Cycle 1
MUL R1, R2 → R3	F
ADD R3, R4 → R5	
ADD R2, R6 → R7	
ADD R8, R9 → R10	
MUL R7, R10 → R11	
ADD R5, R11 → R5	

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		3
R4	1		4
R5	1		5
R6	1		6
R7	1		7
R8	1		8
R9	1		9
R10	1		10
R11	1		11

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a						
b						
c						
d						



	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x						
y						
z						
t						



Cycle 2

MUL gets decoded and allocated into RS x

Step 1: Check if reservation station available. Yes: x

Step 2: Access *the Register Alias Table*

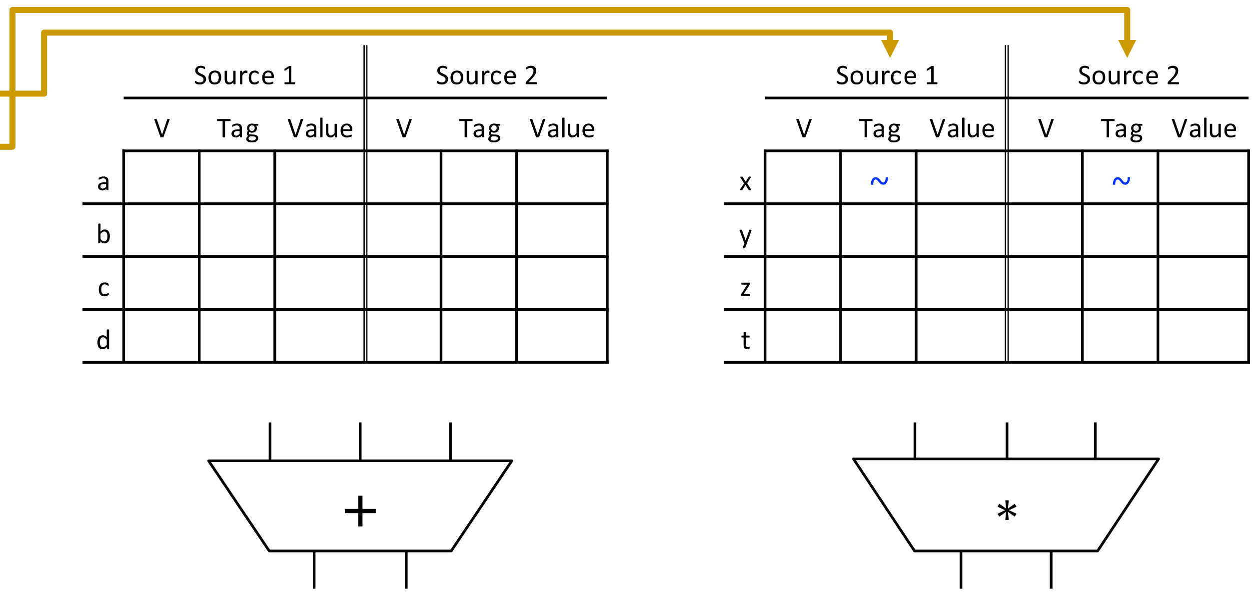
Step 3: Put source registers into reservation station x

Step 4: Rename destination register R3 → x

R3 is now renamed to x.
Its new value will produced by the *reservation station* that is identified by tag x.

Cycle	1	2
`MUL R1, R2 → R3`	F	D
`ADD R3, R4 → R5`		F
`ADD R2, R6 → R7`		
`ADD R8, R9 → R10`		
`MUL R7, R10 → R11`		
`ADD R5, R11 → R5`		

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	0	x	
R4	1		4
R5	1		5
R6	1		6
R7	1		7
R8	1		8
R9	1		9
R10	1		10
R11	1		11



MUL in RS x is ready to execute in the next cycle!

Cycle 3

1. MUL in RS x starts executing

2. ADD gets decoded and allocated into RS a

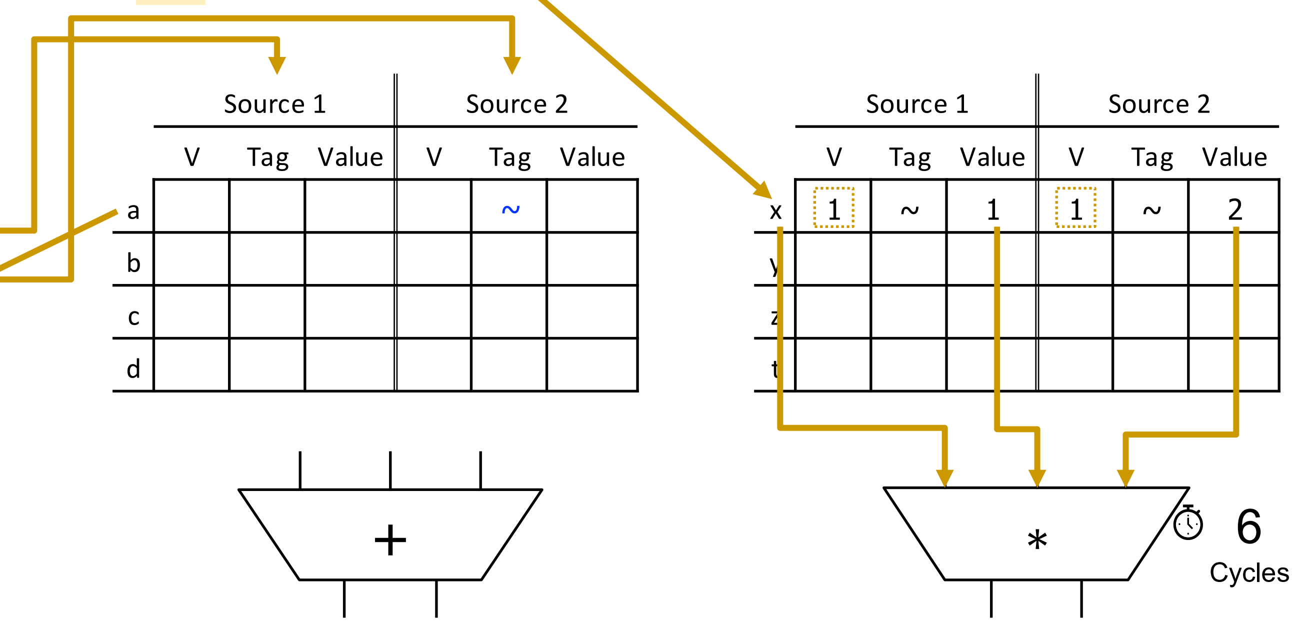
Check readiness (Both sources ready?) → Wakeup

Ready → Dispatch the instruction to the MUL unit

Same Steps 1-4 for ADD… Rename R5 → a

Cycle	1	2	3
MUL R1, R2 → R3	F	D	E_1
ADD R3, R4 → R5		F	D
ADD R2, R6 → R7			F
ADD R8, R9 → R10			
MUL R7, R10 → R11			
ADD R5, R11 → R5			

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	0	x	
R4	1		4
R5	0	a	
R6	1		6
R7	1		7
R8	1		8
R9	1		9
R10	1		10
R11	1		11



ADD in *RS a* cannot execute in the next cycle: one source is not valid

Cycle 4

Cycle	1	2	3	4
MUL R1, R2 → R3	F	D	E_1	E_2
ADD R3, R4 → R5		F	D	-
ADD R2, R6 → R7			F	D
ADD R8, R9 → R10				F
MUL R7, R10 → R11				
ADD R5, R11 → R5				

ADD in RS a waits because one source is not valid.

Rename R7 → b

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	0	x	
R4	1		4
R5	0	a	
R6	1		6
R7	0	b	
R8	1		8
R9	1		9
R10	1		10
R11	1		11

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	0	x		1	~	4
b		~			~	
c						
d						

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y						
z						
t						

ADD in RS b is ready to execute in the next cycle!

It will be executed out of order in the next cycle.

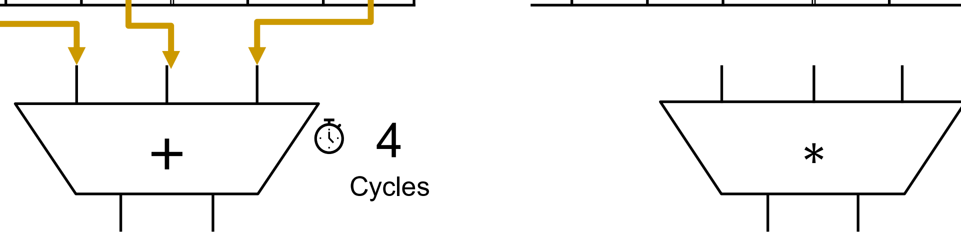
Cycle 5

Instruction	1	2	3	4	5
MUL R1, R2 → R3	F	D	E_1	E_2	E_3
ADD R3, R4 → R5		F	D	-	-
ADD R2, R6 → R7			F	D	E_1
ADD R8, R9 → R10				F	D
MUL R7, R10 → R11					F
ADD R5, R11 → R5					

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	0	x	
R4	1		4
R5	0	a	
R6	1		6
R7	0	b	
R8	1		8
R9	1		9
R10	0	c	
R11	1		11

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	0	x		1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d						

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y						
z						
t						



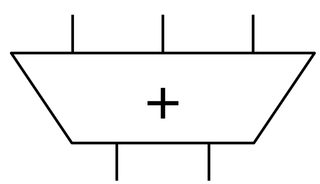
ADD in RS c is ready to execute in the next cycle!

Cycle 6

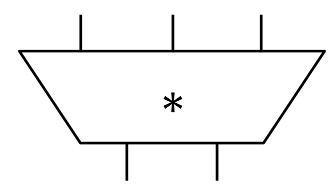
Instruction	1	2	3	4	5	6
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4
ADD R3, R4 → R5		F	D	-	-	-
ADD R2, R6 → R7			F	D	E_1	E_2
ADD R8, R9 → R10				F	D	E_1
MUL R7, R10 → R11					F	D
ADD R5, R11 → R5						F

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	0	x	
R4	1		4
R5	0	a	
R6	1		6
R7	0	b	
R8	1		8
R9	1		9
R10	0	c	
R11	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	0	x		1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d						



	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	0	b		0	c	
z						
t						



Cycle 7

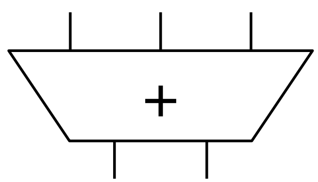
All six instructions are now decoded and renamed

Note what happened to R5: Renamed twice!

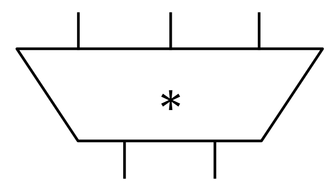
Cycle	1	2	3	4	5	6	7
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5
ADD R3, R4 → R5		F	D	-	-	-	-
ADD R2, R6 → R7			F	D	E_1	E_2	E_3
ADD R8, R9 → R10				F	D	E_1	E_2
MUL R7, R10 → R11					F	D	-
ADD R5, R11 → R5						F	D

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	0	x	
R4	1		4
R5	0	d	
R6	1		6
R7	0	b	
R8	1		8
R9	1		9
R10	0	c	
R11	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	0	x		1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	0	a		0	y	



	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	0	b		0	c	
z						
t						



Cycle 8 (First Slide)

Instruction	1	2	3	4	5	6	7	8
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6
ADD R3, R4 → R5		F	D	-	-	-	-	
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	
ADD R8, R9 → R10				F	D	E_1	E_2	
MUL R7, R10 → R11					F	D	-	
ADD R5, R11 → R5						F	D	

MUL in RS x is done

Broadcast MUL's tag (x)

- ✓ Check tag
- ✓ Check for invalidity

Broadcast MUL's result (2)

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	0	b	
R8	1		8
R9	1		9
R10	0	c	
R11	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1		2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	0	a		0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	0	b		0	c	
z						
t						

ADD in RS a is ready to execute in the next cycle!

Cycle 8 (Second Slide)

Instruction	Cycle 1	2	3	4	5	6	7	8
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6
ADD R3, R4 → R5		F	D	-	-	-	-	-
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4
ADD R8, R9 → R10				F	D	E_1	E_2	
MUL R7, R10 → R11					F	D	-	
ADD R5, R11 → R5						F	D	

ADD in RS b is also done

Broadcast ADD's tag (b)

✓ Check tag
✓ Check for invalidity

Broadcast ADD's result (8)

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	0	c	
R11	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	0	a		0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1		8	0	c	
z						
t						

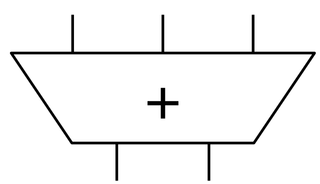
MUL in RS y is still NOT ready to execute in the next cycle!

Cycle 8 (Third Slide)

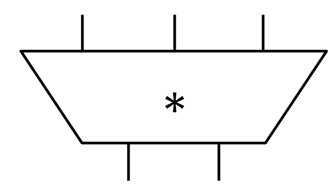
	Cycle	1	2	3	4	5	6	7	8
MUL R1, R2 → R3		F	D	E_1	E_2	E_3	E_4	E_5	E_6
ADD R3, R4 → R5			F	D	-	-	-	-	-
ADD R2, R6 → R7				F	D	E_1	E_2	E_3	E_4
ADD R8, R9 → R10					F	D	E_1	E_2	E_3
MUL R7, R10 → R11						F	D	-	-
ADD R5, R11 → R5							F	D	-

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	0	c	
R11	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	0	a		0	y	



	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	0	c	
z						
t						



Cycle 9

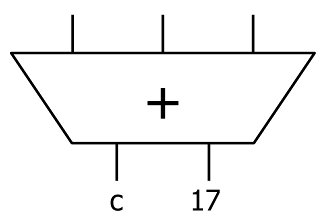
	Cycle	1	2	3	4	5	6	7	8	9
MUL R1, R2 → R3		F	D	E_1	E_2	E_3	E_4	E_5	E_6	W
ADD R3, R4 → R5			F	D	-	-	-	-	-	E_1
ADD R2, R6 → R7				F	D	E_1	E_2	E_3	E_4	W
ADD R8, R9 → R10					F	D	E_1	E_2	E_3	E_4
MUL R7, R10 → R11						F	D	-	-	-
ADD R5, R11 → R5							F	D	-	-

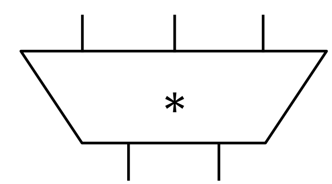
Broadcast and Update

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	0	a		0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						





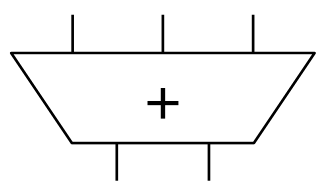
MUL in RS y is ready to execute in the next cycle!

Cycle 10

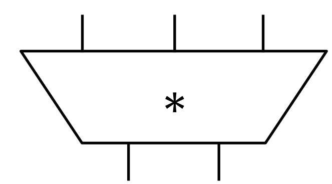
Instruction	1	2	3	4	5	6	7	8	9	10
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6	W	
ADD R3, R4 → R5		F	D	-	-	-	-	-	E_1	E_2
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4	W	
ADD R8, R9 → R10				F	D	E_1	E_2	E_3	E_4	W
MUL R7, R10 → R11					F	D	-	-	-	E_1
ADD R5, R11 → R5						F	D	-	-	-

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	0	a		0	y	



	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						

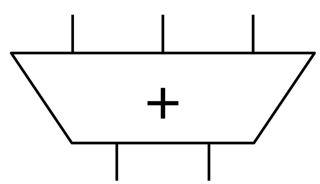


Cycle 11

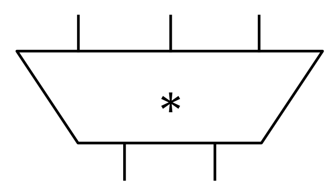
Instruction	1	2	3	4	5	6	7	8	9	10	11
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6	W		
ADD R3, R4 → R5		F	D	-	-	-	-	-	E_1	E_2	E_3
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4	W		
ADD R8, R9 → R10				F	D	E_1	E_2	E_3	E_4	W	
MUL R7, R10 → R11					F	D	-	-	-	E_1	E_2
ADD R5, R11 → R5						F	D	-	-	-	-

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	0	a		0	y	



	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						



Cycle 12

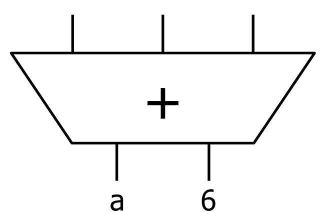
Instruction	1	2	3	4	5	6	7	8	9	10	11	12
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6	W			
ADD R3, R4 → R5		F	D	-	-	-	-	-	E_1	E_2	E_3	E_4
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4	W			
ADD R8, R9 → R10				F	D	E_1	E_2	E_3	E_4	W		
MUL R7, R10 → R11					F	D	-	-	-	E_1	E_2	E_3
ADD R5, R11 → R5						F	D	-	-	-	-	-

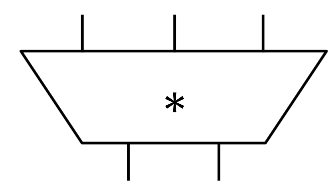
Broadcast and Update

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	1	~	6	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						



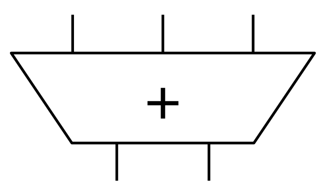


Cycle 13

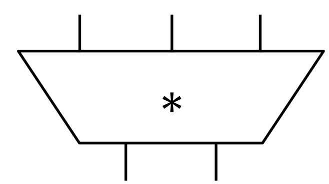
Instruction	1	2	3	4	5	6	7	8	9	10	11	12	13
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6	W				
ADD R3, R4 → R5		F	D	-	-	-	-	-	E_1	E_2	E_3	E_4	W
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4	W				
ADD R8, R9 → R10				F	D	E_1	E_2	E_3	E_4	W			
MUL R7, R10 → R11					F	D	-	-	-	E_1	E_2	E_3	E_4
ADD R5, R11 → R5						F	D	-	-	-	-	-	-

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	1	~	6	0	y	



	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						

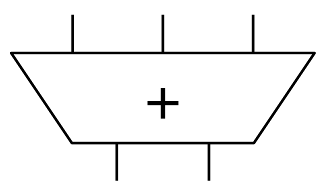


Cycle 14

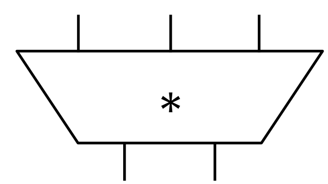
Instruction	1	2	3	4	5	6	7	8	9	10	11	12	13	14
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6	W					
ADD R3, R4 → R5		F	D	-	-	-	-	-	E_1	E_2	E_3	E_4	W	
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4	W					
ADD R8, R9 → R10				F	D	E_1	E_2	E_3	E_4	W				
MUL R7, R10 → R11					F	D	-	-	-	E_1	E_2	E_3	E_4	E_5
ADD R5, R11 → R5						F	D	-	-	-	-	-	-	-

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	0	y	

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	1	~	6	0	y	



	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						



Cycle 15

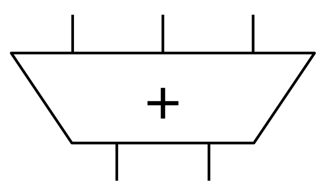
Instruction	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6	W						
ADD R3, R4 → R5		F	D	-	-	-	-	-	E_1	E_2	E_3	E_4	W		
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4	W						
ADD R8, R9 → R10				F	D	E_1	E_2	E_3	E_4	W					
MUL R7, R10 → R11					F	D	-	-	-	E_1	E_2	E_3	E_4	E_5	E_6
ADD R5, R11 → R5						F	D	-	-	-	-	-	-	-	-

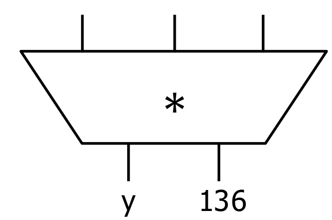
Broadcast and Update

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	1		136

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	1	~	6	1	~	136

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						





ADD in RS d is ready to execute in the next cycle!

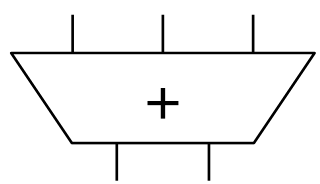
Cycle 16

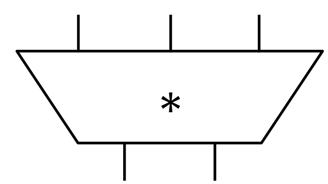
Cycle	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6	W							
ADD R3, R4 → R5		F	D	-	-	-	-	-	E_1	E_2	E_3	E_4	W			
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4	W							
ADD R8, R9 → R10				F	D	E_1	E_2	E_3	E_4	W						
MUL R7, R10 → R11					F	D	-	-	-	E_1	E_2	E_3	E_4	E_5	E_6	W
ADD R5, R11 → R5						F	D	-	-	-	-	-	-	-	-	E_1

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	1		136

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	1	~	6	1	~	136

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						





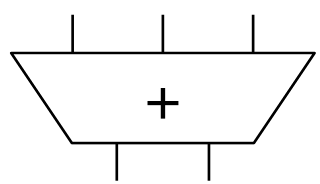
Cycle 17

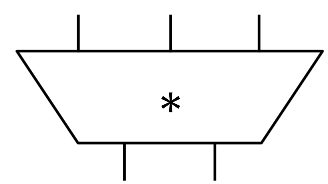
Instruction	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6	W								
ADD R3, R4 → R5		F	D	-	-	-	-	-	E_1	E_2	E_3	E_4	W				
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4	W								
ADD R8, R9 → R10				F	D	E_1	E_2	E_3	E_4	W							
MUL R7, R10 → R11					F	D	-	-	-	E_1	E_2	E_3	E_4	E_5	E_6	W	
ADD R5, R11 → R5						F	D	-	-	-	-	-	-	-	-	E_1	E_2

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	1		136

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	1	~	6	1	~	136

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						



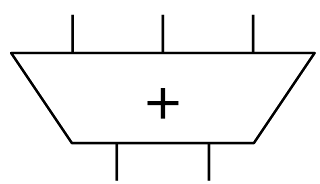


Cycle 18

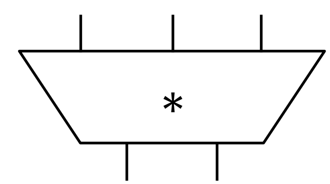
Instruction	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6	W									
ADD R3, R4 → R5		F	D	-	-	-	-	-	E_1	E_2	E_3	E_4	W					
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4	W									
ADD R8, R9 → R10				F	D	E_1	E_2	E_3	E_4	W								
MUL R7, R10 → R11					F	D	-	-	-	E_1	E_2	E_3	E_4	E_5	E_6	W		
ADD R5, R11 → R5						F	D	-	-	-	-	-	-	-	-	E_1	E_2	E_3

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	0	d	
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	1		136

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	1	~	6	1	~	136



	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						



Cycle 19

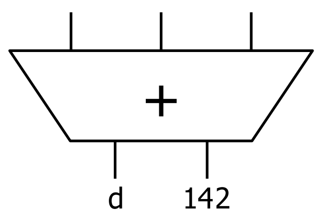
Instruction	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6	W										
ADD R3, R4 → R5		F	D	-	-	-	-	-	E_1	E_2	E_3	E_4	W						
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4	W										
ADD R8, R9 → R10				F	D	E_1	E_2	E_3	E_4	W									
MUL R7, R10 → R11					F	D	-	-	-	E_1	E_2	E_3	E_4	E_5	E_6	W			
ADD R5, R11 → R5						F	D	-	-	-	-	-	-	-	-	E_1	E_2	E_3	E_4

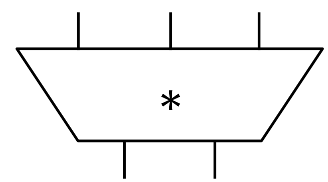
Broadcast and Update

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	1		142
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	1		136

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	1	~	6	1	~	136

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						



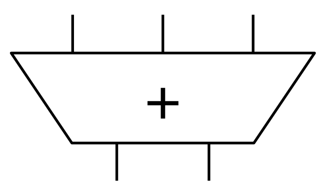


Cycle 20

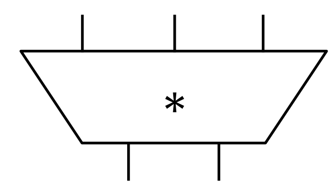
Instruction	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
MUL R1, R2 → R3	F	D	E_1	E_2	E_3	E_4	E_5	E_6	W											
ADD R3, R4 → R5		F	D	-	-	-	-	-	E_1	E_2	E_3	E_4	W							
ADD R2, R6 → R7			F	D	E_1	E_2	E_3	E_4	W											
ADD R8, R9 → R10				F	D	E_1	E_2	E_3	E_4	W										
MUL R7, R10 → R11					F	D	-	-	-	E_1	E_2	E_3	E_4	E_5	E_6	W				
ADD R5, R11 → R5						F	D	-	-	-	-	-	-	-	-	E_1	E_2	E_3	E_4	W

Register	Valid	Tag	Value
R1	1		1
R2	1		2
R3	1		2
R4	1		4
R5	1		142
R6	1		6
R7	1		8
R8	1		8
R9	1		9
R10	1		17
R11	1		136

	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
a	1	~	2	1	~	4
b	1	~	2	1	~	6
c	1	~	8	1	~	9
d	1	~	6	1	~	136



	Source 1			Source 2		
	V	Tag	Value	V	Tag	Value
x	1	~	1	1	~	2
y	1	~	8	1	~	17
z						
t						



Some Questions

- What is needed in hardware to perform tag broadcast and value capture?

  → make a value valid

  → wake up an instruction

  Wires, Comparators & Logic

- Does the tag have to be the ID of the Reservation Station Entry?

  No, could be **any unique name that enables linking of producer to consumer**

- What can potentially become the critical path?
  - **Tag broadcast → value capture → instruction wake up**

- How can you reduce the potential critical paths?
  - **More pipelining and prediction**